Measuring the Cost of Xn Handover in Non-Terrestrial Networks: An Emulation Study Driven by Real LEO Ephemeris

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***Abstract*—Handover studies for non-terrestrial networks (NTN) almost universally count handovers, treating each as a cost-free event. This obscures a term that matters: handovers differ substantially in what they cost, and the policies that minimise handover *count* systematically select the most expensive ones. We present an emulation testbed in which real 3GPP Xn handovers are executed between software gNBs while propagation delays derived from actual Starlink ephemeris are injected on individual 5G interfaces. Across 470 measured handovers we decompose handover latency into its signalling and core-processing components and quantify, for the first time to our knowledge, how much user-plane interruption a millisecond of delay on each link actually buys. We find a strong asymmetry: delay on the inter-satellite Xn link inflates handover latency ($+0.46$ ms/ms) but contributes essentially *zero* interruption (-0.01 ms/ms), because it is paid before the UE leaves the source cell, whereas feeder- and service-link delay convert almost directly into downtime ($+0.80$ and $+1.24$ ms/ms). Replaying real satellite pass sequences, we show that a maximum-elevation policy performs $7\times$ more handovers than a maximum-service-time policy but each costs only half as much (9.55 vs. 19.47 ms), so the $7\times$ count advantage reduces to $3.4\times$ in cumulative interruption. Counting handovers therefore overstates the benefit of stability-focused policies, because the handovers they avoid are the cheap ones.**

***Index Terms*—non-terrestrial networks, LEO satellite, Xn handover, 5G, handover interruption, mobility management, testbed**

I. Introduction

Low Earth Orbit (LEO) constellations move fast enough that a satellite serves a given ground user for only a few minutes, making mobility management a dominant concern in non-terrestrial networks. A large literature accordingly compares satellite-selection policies—maximum elevation, maximum remaining service time, maximum SINR, and hybrids—and reports how many handovers each incurs [5]–[8].

That body of work shares a modelling assumption: a handover is an instantaneous event whose only cost is that it happened. In a real 5G system a handover is a multi-message procedure spanning the radio interface, the inter-base-station interface, and the core network, and its duration depends on the propagation delay of the links it traverses. Under NTN geometry those delays vary by more than a factor of three across a single satellite pass. A policy that chooses *when* to hand over is therefore also choosing *how expensive* that handover will be—a coupling that count-based studies cannot express.

This paper measures that coupling. Our contributions are:

- An emulation testbed that performs standards-conformant 3GPP Xn handovers between software gNBs, with per-link propagation delay injected from real Starlink ephemeris (Sections III–IV).
- A decomposition of handover cost showing that *where* delay sits determines whether it causes interruption. Xn-link delay costs latency but not downtime; feeder- and service-link delay cost both (Section VI-B).
- A live replay of real pass sequences establishing that per-handover cost is inversely related to handover frequency, which materially changes the policy trade-off reported in the literature (Section VI-D).

II. Background

A. Xn Handover in 3GPP

In an Xn handover the source and target gNBs communicate directly rather than routing the procedure through the core [1]. The sequence is: (i) the source sends a *Handover Request* carrying the UE context over Xn; (ii) the target admits the UE and acknowledges; (iii) the source sends the UE an `RRCReconfiguration` containing `reconfigurationWithSync`—the handover command [2]; (iv) the UE synchronises to the target and replies `RRCReconfigurationComplete`; (v) the target sends a *Path Switch Request* to the AMF, which redirects the downlink tunnel via the SMF and UPF; (vi) the target releases the source context.

Two properties matter here. The path switch occurs *after* the UE has arrived, not before, so the core is only asked to move the tunnel once the UE is reachable at the target. And for an intra-AMF, intra-tracking-area handover the UE remains RRC-CONNECTED throughout, so no NAS signalling is generated.

Baseline 3GPP handover is *break-before-make*: the UE stops using the source before it starts using the target, and a short user-plane gap is expected. Reference interruption times are $\geq$45 ms for LTE and 24 ms (FR1) / 7.4 ms (FR2) for NR with a make-before-break configuration [3]. Dual Active Protocol Stack (DAPS) handover, introduced in Rel-16, targets near-zero interruption but is *explicitly not supported for NTN*,

TABLE I
TESTBED CONFIGURATION

Component	Configuration
Core network	Open5GS (AMF, SMF, UPF, and support NFs)
RAN	UERANSIM with Xn handover extension
gNBs	3, common PLMN / TAC / S-NSSAI, one AMF
UE	1, single PDU session, isolated namespace
Delay injection	per-interface, per-gNB
Ephemeris	Starlink TLE, Montreal ground station

as maintaining simultaneous connections to two satellites is impractical given payload and feeder-link constraints [4]. NTN instead relies on RACH-less handover and conditional handover with NTN-specific triggers.

B. Mobility Triggers in NTN

3GPP defines no elevation-angle event. The NTN-specific Rel-17 conditional handover conditions are CONDEVENT D1 (distance to the serving cell reference location above one threshold and distance to the candidate below another) and CONDEVENT T1 (a time window) [4]. Because elevation angle and slant range are monotonically related, an elevation threshold is equivalent to a D1-style distance condition, and a remaining-service-time criterion is a T1-style time condition. We use elevation throughout for interpretability while noting this correspondence.

We emphasise that the mechanism studied here is *elevation-triggered handover*, not conditional handover: the network evaluates the condition and issues a command. In true CHO the UE holds candidate configurations and evaluates the conditions autonomously.

III. TESTBED

A. Architecture

The testbed comprises an Open5GS [11] 5G core and a modified UERANSIM [12] radio-access simulator, with three software gNBs and one UE. All gNBs register to the same AMF with a common tracking area and slice identifier, since an Xn path switch is only meaningful when source and target are served by the same AMF. Table I summarises the configuration.

B. Xn Handover Procedure

UERANSIM does not natively implement inter-gNB mobility. We added a complete Xn handover following the 3GPP ordering described above. The source gNB transfers the UE context—security parameters, PDU session tunnel endpoints and QoS flow identifiers—to the target over a dedicated Xn interface. The target admits the UE and holds the context in a pending state. The source then issues a genuine `RRCReconfiguration` carrying `reconfigurationWithSync`, which identifies the target cell; it does *not* release the UE. On receiving the command the UE re-points its serving cell, remains RRC-CONNECTED, and replies `RRCReconfigurationComplete` to the target. Only then does the target issue the NGAP *Path Switch Request*, and only on acknowledgement does it instantiate the user-plane tunnels.

Three design points are worth stating because they determine what the measurements mean. First, the path switch is deferred until the UE has actually arrived, so a handover the UE never completes never reaches the core. Second, source and target correlate the UE through an identifier the UE itself generates and both gNBs already possess, so the context created by the path switch *is* the context the UE is connected through—there is no second, orphaned context. Third, the UE never enters idle, so no NAS re-registration occurs; this is verified in every run by the absence of registration signalling.

We do not implement SN Status Transfer or a data-forwarding tunnel, so the procedure is not lossless; a brief user-plane gap is expected and is precisely the quantity we measure.

C. Isolation of the Data Path

A subtlety with material consequences: when the UE and the core share a network namespace, the UE's IP address is local to the host, and the kernel's local routing table short-circuits every downlink packet to a local socket. Downlink then never traverses the GTP tunnel, and any user-plane measurement is meaningless while appearing to succeed. We therefore run the UE in an isolated network namespace, which forces downlink through the full UPF $\rightarrow$ GTP $\rightarrow$ gNB $\rightarrow$ radio path. The distinction is visible in round-trip time: <0.1 ms when short-circuited versus 1–6 ms through the real path.

IV. NTN MODELLING

A. Geometry

Satellite positions are propagated with SGP4 [13] from real Starlink two-line elements over a ground station at Montreal (45.5017° N, 73.5673° W). Two choices preserve fidelity. Passes are computed at the TLE *epoch* rather than at wall-clock time, since a TLE is accurate only for a few days around its epoch. And only satellites within the 500–600 km operational shell are considered, because the element set also contains lowered and de-orbiting satellites whose geometry would understate a representative LEO link.

The selected reference pass (STARLINK-3075, 87° culmination) spans slant ranges of 544–1805 km, giving one-way propagation delays of 1.82–6.02 ms. Fig. 1 shows delay against elevation with the sweep points marked. Over a one-hour window the constellation presents 270 passes with a median duration of 7.4 minutes and up to 43 satellites simultaneously visible above 10°.

B. Delay Injection and Architecture Scenarios

Propagation delay is injected independently on each 5G interface using kernel traffic shaping, applied at both ends of every link so that each side carries the one-way delay and the round trip emerges naturally. Jitter is deliberately not used: it reorders packets, which propagation delay does not, and would corrupt the interruption measurement.

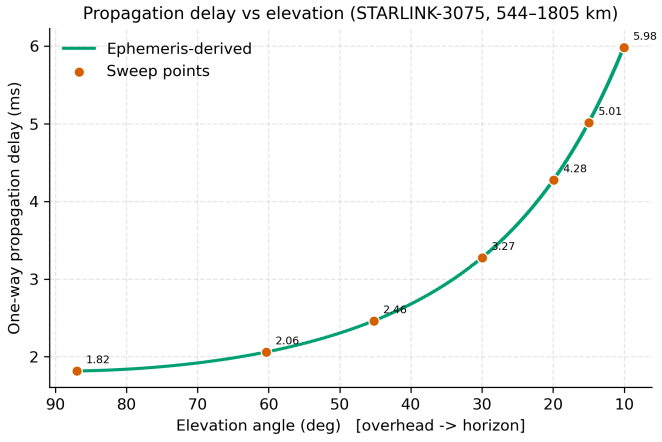


Fig. 1. One-way propagation delay against elevation angle, derived from real Starlink ephemeris over Montreal. Markers indicate the operating points used in the delay sweep.

TABLE II
NTN ARCHITECTURE SCENARIOS

Scenario	gNB location	Long link
Transparent	Ground	Radio (service + feeder)
Regenerative	On satellite	Feeder (N2 + N3)
Inter-satellite	Two satellites	Xn

Which interface carries the long delay depends on payload architecture, giving three scenarios (Table II). Under a *transparent* payload the satellite is a bent-pipe relay and the gNB is on the ground, so the service and feeder hops together form the radio link. Under a *regenerative* payload the gNB is aboard the satellite, so the feeder link to the ground carries N2 and N3. When source and target gNBs are on different satellites, the *inter-satellite* link carries Xn.

C. Handover Policies

Three satellite-selection policies from the LEO literature are implemented. *Maximum elevation* always serves on the highest-elevation visible satellite. *Elevation threshold* holds the serving satellite until its elevation falls below a threshold. *Maximum remaining service time* holds until the serving satellite is close to setting, then selects the candidate with the longest remaining visibility.

All three apply the stability controls that make a trigger usable in practice: a hysteresis margin, so a candidate must exceed the serving cell by a margin rather than merely exceed it; a time-to-trigger, so the condition must hold continuously; a minimum dwell time between handovers; and a lead time, so the handover completes before the serving link becomes unusable. The lead time is the direct coupling between measured handover cost and policy design.

V. EXPERIMENTAL DESIGN

A. Metrics

Each handover yields three latency components measured by the target gNB with microsecond-resolution timestamps:

- **UE arrival** — from the Xn Handover Request arriving at the target to the UE completing its move. Covers Xn signalling, handover command delivery, and the radio switch.
- **Path switch** — from Path Switch Request to Acknowledge. This is core processing: AMF, SMF and UPF updating the tunnel.
- **Total** — the whole procedure at the target.

User-plane interruption is measured with a high-rate probe traversing the GTP tunnel throughout each handover. The outage is computed as the number of *contiguous* missing probes multiplied by the inter-arrival spacing *actually achieved*. Both halves matter: the contiguous count isolates the handover from unrelated scattered loss, while the achieved spacing is required because probe pacing degrades toward the round-trip rate once delay is injected. Measurement resolution is consequently one probe interval, 1.2 ms undelayed and 5–8 ms under injected delay.

B. Experimental Controls

Two effects were identified during validation that would otherwise confound the results, and both are controlled.

Handover cadence affects core processing. At 2.2 s between handovers the path-switch time nearly doubles relative to 5 s, as the core does not settle between procedures. Every configuration therefore uses a fixed cycle period so the core sees an identical handover rate throughout.

Probe traffic perturbs the measurement. At 1000 probes per second the measurement traffic queues behind the delayed link and inflates the measured path-switch time. This is quantified in Section VI-B and bounds the precision of one reported coefficient.

Each measurement session is bracketed by identical baseline runs to detect drift. Across a full session the baseline moved from 3.897 ms to 4.087 ms median path-switch time, confirming stability.

C. Experiments

E1 establishes the zero-delay baseline (50 handovers). **E2** sweeps the seven elevation operating points for each of the three architecture scenarios, 20 handovers per point (420 total). **E3** evaluates the three policies offline against the real constellation schedule, isolating decision behaviour from execution cost. **E4** replays real pass sequences against the live testbed: each gNB's service-link delay tracks whichever satellite it currently represents, and a real Xn handover fires when the policy dictates. Four independent 15-minute windows are used per policy. Because only two gNBs are needed—the freed gNB is reassigned to the next satellite after each handover—arbitrarily long pass chains are supported. Geometry is replayed on an accelerated clock, which is sound because geometry evolves over minutes while a handover completes in milliseconds; the handovers themselves execute in real time.

TABLE III
SENSITIVITY TO INJECTED LINK RTT (MS PER MS), $n = 140$ PER SCENARIO

Scenario	UE arrival	Path switch	Total	Interruption
Regenerative	$+0.06$	$+1.36$	$+1.42$	$+0.80$
Transparent	$+1.16$	$+0.35$	$+1.50$	$+1.24$
Inter-satellite	$+0.48$	-0.03	$+0.46$	-0.01

TABLE IV
MEASURED POLICY COST, MEAN ± S.D. OVER FOUR INDEPENDENT WINDOWS

Policy	HO per window	Cost per HO (ms)	Cumulative outage (ms)	Mean link delay (ms)
Max. elevation	10.5	9.55 ± 1.82	97.5 ± 16.7	2.15
Elev. threshold	1.5	19.21 ± 1.97	26.4 ± 12.2	5.22
Max. service time	1.5	19.47 ± 1.08	29.1 ± 16.7	5.33

VI. RESULTS

A. Baseline

Over 50 handovers with no injected delay the procedure completes in a median of 4.744 ms (mean 4.891 ± 0.664 ms), decomposing into 0.831 ms of UE arrival and 3.897 ms of path switch. The core round trip therefore dominates, accounting for 80% of the total. All three distributions are tight, with 95th percentiles within 10% of their medians (1.296, 4.418 and 5.384 ms respectively). Measured interruption is 1.24 ms, at the measurement resolution.

Every run confirms the procedure is a true handover: no RRC release, no transition to idle, no NAS re-registration, and exactly one UE context at the target.

B. Where Delay Costs What

Table III reports the sensitivity of each metric to injected link round-trip time, and Fig. 2 shows the underlying curves.

The results confirm the expected structure and isolate the asymmetry that motivates this work.

Regenerative UE-arrival sensitivity is $+0.06$, i.e. flat: feeder delay does not touch Xn signalling or the radio interface. Transparent UE-arrival sensitivity is $+1.16 \approx 1$, consistent with the handover command traversing the source radio link and the completion message traversing the target link—one round trip in total. Inter-satellite path-switch sensitivity is -0.03, i.e. zero: Xn delay genuinely does not touch the core round trip.

The inter-satellite UE-arrival slope of $+0.48$ deserves comment because it appears wrong. One would expect 1.0 for an Xn round trip. It is correct: the measurement window opens when the target *receives* the Handover Request, so the request's own flight time falls outside it and only the acknowledgement direction—one one-way hop, half a round trip—is captured. Obtaining 0.48 rather than 1.0 confirms the instrumentation measures the interval it claims.

The decisive result is the interruption column. Inter-satellite delay produces -0.01 ms of interruption per millisecond of link RTT: it is entirely free in downtime terms, because the Xn exchange completes *before* the UE is instructed to move. Regenerative and transparent delay produce $+0.80$ and $+1.24$ respectively, because they are incurred while the UE is between cells. The same millisecond of propagation delay therefore costs radically different amounts depending on which link carries it—a distinction invisible to count-based analyses and directly actionable in constellation design and satellite-selection optimisation.

One caveat bounds the transparent path-switch figure. Radio delay cannot physically affect the gNB–AMF round trip, yet $+0.35$ is measured. Repeating the scenario without probe traffic yields -0.11, identifying this as measurement cross-talk rather than a property of the system.

C. Policy Behaviour

Evaluated offline against the real constellation schedule, the three policies reproduce the trade-off documented in the literature. Maximum elevation performs 8 handovers per 15-minute window while maintaining a mean serving elevation of $52.7°$ (2.58 ms mean delay). Both stability-focused policies perform 2, at $25.9°$ and $24.0°$ respectively. The elevation-threshold policy additionally produces a *late* handover, in which the serving satellite is lost before the switch fires—the known weakness of a fixed threshold, since two passes may share an elevation while having very different remaining service time. The maximum-service-time policy avoids this at equal handover count.

The resulting serving-elevation trajectories differ sharply. Maximum elevation produces a characteristic sawtooth, abandoning satellites after a median of 1.3 minutes of a 7.4-minute pass—discarding roughly 80% of each satellite's usable service time—while the stability-focused policies retain each satellite for approximately 6 minutes. These trajectories are visible per policy in Fig. 4.

D. Measured Cost of Policy Choice

E4 executes those decisions against the live testbed, so each handover carries its measured cost. Table IV reports results over four independent 15-minute windows.

The central finding is in the second column. Handovers under the maximum-service-time policy cost *twice as much each* as those under maximum elevation (19.47 vs. 9.55 ms). The reason is structural: a policy that defers handover until the serving satellite is nearly setting necessarily executes every handover across the longest, slowest links available. The policy that minimises handover *count* maximises handover *cost*.

This materially changes the trade-off. On counts alone, maximum service time appears $7\times$ better (1.5 vs. 10.5 handovers). In measured cumulative interruption the advantage is only $3.4\times$ (29.1 vs. 97.5 ms). Roughly half the apparent benefit is consumed by the higher per-event cost. Fig. 3 presents this as the frontier an optimiser must trade along, and Fig. 4 shows the underlying pass sequences with per-handover costs.

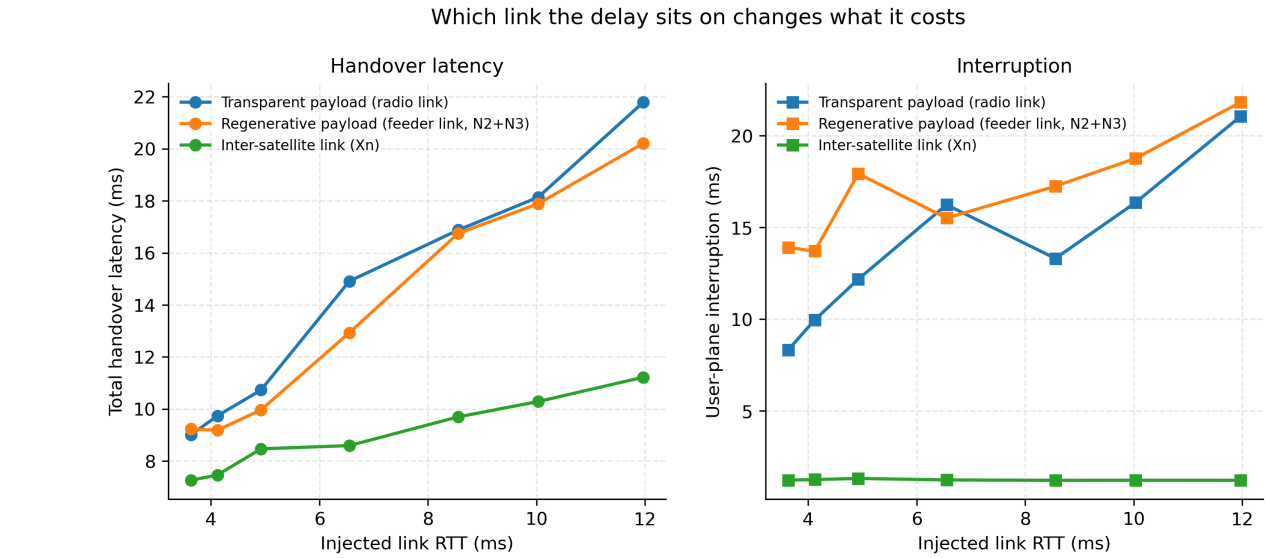


Fig. 2. Handover latency (left) and user-plane interruption (right) against injected link round-trip time. All architectures pay latency, but only those whose delay is incurred after the UE leaves the source cell pay interruption.



Fig. 3. Measured trade-off between link quality and cumulative user-plane interruption. Error bars span four independent 15-minute windows.

Fig. 5 makes the mechanism explicit: handover cost rises as the elevation of the departing satellite falls, because both the source and target links are longer.

Repeating an identical window yields 78.5 versus 78.0 ms cumulative cost for maximum elevation (0.6% difference) and 36.5 versus 37.0 ms for the elevation threshold (1.4%), confirming reproducibility where the sample is adequate.

VII. Comparison with the Literature

Our geometry agrees with published values: LEO one-way propagation delay is reported as 1.6–6 ms, against our ephemeris-derived 1.82–6.02 ms. The observation that a LEO satellite serves a user for only a few minutes is consistent with our 7.4-minute median pass [9], [10].

For policy behaviour the closest published comparison evaluates the same two strategy families over 50 Monte Carlo runs [5], reporting 309.2 handovers for highest-elevation against 146.4 for longest-visibility—a 53% reduction—at a cost of 3.7 dB in mean SINR. We observe an 86% reduction at a cost of 2.4× in mean link delay. The direction, mechanism and qualitative conclusion agree; our larger reduction is expected because our selection is purely geometric, whereas SINR incorporates fading and antenna effects that partly decouple from geometry. Related studies report reductions of 40–81% [6], [7], placing our figure just above a well-populated range. That work states the trade-off explicitly—strategies maximising instantaneous link quality trigger excessive handovers, while stability-focused approaches sacrifice link performance—which is the frontier we measure.

Our interruption values (8–22 ms) are below published handover interruption times (∼35 ms intra-gNB, ≥45 ms LTE, 24/7.4 ms NR FR1/FR2). This is by construction, not by improvement: the testbed has no physical layer, and the literature attributes 10 ms to UE RRC processing alone. Our figures should be read as the *network-side contribution*, with radio-layer terms excluded.

What we did not find in the literature is any measurement of *which link's* delay converts into interruption, nor of per-handover cost as a function of policy. Both follow from the 3GPP procedure ordering and are therefore unsurprising once stated; the contribution is their quantification.

VIII. Limitations

The testbed has no physical layer: no random access, no timing advance, no Doppler, fading or beam modelling. Absolute interruption times are consequently lower bounds representing signalling and core processing only. A useful consequence is that the emulated handover is inherently RACH-less, which

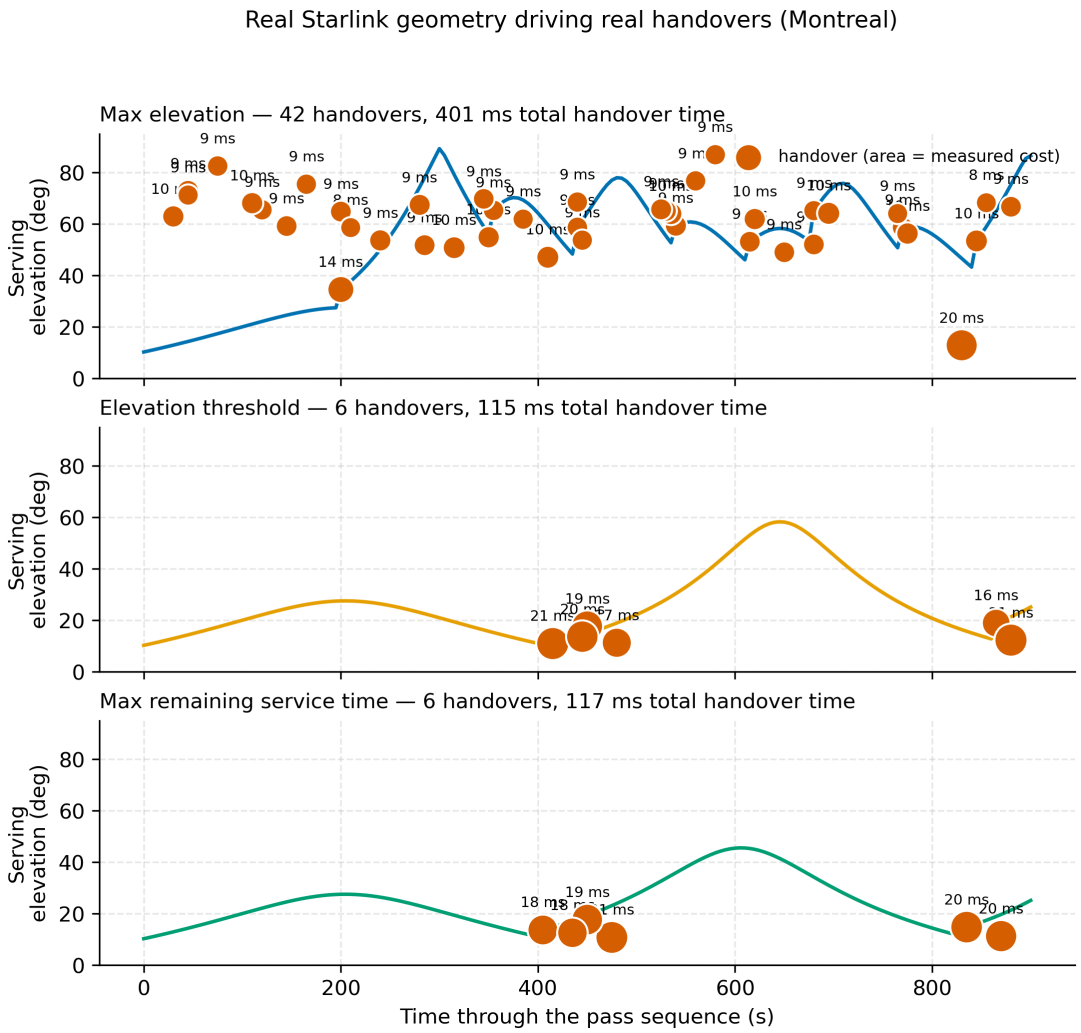


Fig. 4. Real Starlink geometry driving real handovers. Marker area is proportional to measured handover cost; low-elevation handovers are visibly more expensive.

aligns with the NTN model rather than the terrestrial one, but timing-advance pre-compensation cannot be represented.

The policy layer assumes exact geometry knowledge, consistent with the Rel-17 mandate for GNSS-capable NTN UEs, but ephemeris staleness and GNSS error are not modelled. Delay is injected as a constant per link during each handover, which is sound given that geometry evolves over minutes while a handover completes in milliseconds, but does not capture continuous variation within a procedure.

Interruption resolution is one probe interval—1.2 ms undelayed, 5–8 ms under injected delay—because probe pacing degrades toward the round-trip rate. Outages below this cannot be distinguished from zero. Finally, the two stability-focused policies overlap within error on both frontier axes with six handovers each, so we report them as equivalent rather than ranked.

We also note one unresolved observation: for the inter-satellite scenario the measured outage is consistently one probe interval, shorter than the path-switch window would suggest. We hypothesise that the asynchronous serving-cell update briefly preserves the old path, but have not verified this, and therefore make no precise claim about absolute interruption in that scenario.

IX. Conclusion

We measured the cost of 3GPP Xn handover under NTN propagation conditions using an emulation testbed driven by real Starlink ephemeris. Across 470 handovers we showed that delay on the inter-satellite Xn link inflates handover latency but produces no user-plane interruption, because it is incurred before the UE leaves the source cell, whereas feeder- and service-link delay convert almost directly into downtime. Replaying real pass sequences, we showed that per-handover cost is inversely related to handover frequency, so that counting handovers overstates the benefit of stability-focused policies by roughly a factor of two: the handovers such policies avoid are precisely the cheap ones.

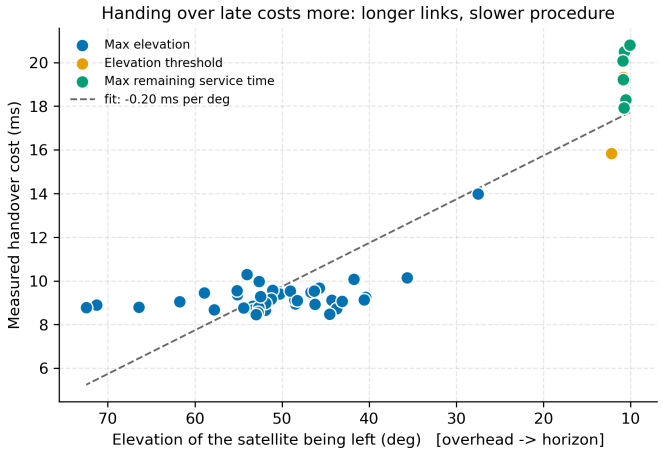


Fig. 5. Measured handover cost against the elevation of the satellite being left. Handing over late is more expensive.

These coefficients are directly usable in constellation and mobility optimisation, where the relevant question is not how many handovers a policy incurs but how much service time they consume. Extending the testbed with a physical-layer model, and with data forwarding for lossless handover, would allow the absolute figures to be compared against over-the-air measurements.

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